Image Deblurring via Hessian Spectral Regularization (Mixed Nuclear + Frobenius)

Outline

- **1:** Blur model (DSP view), why inversion fails, need for regularization
- **2:** Hessian operator, spectral norms, why "nuclear" vs "Frobenius"
- **3:** ADMM splitting + the **only student-coded step**: a proximal operator

Motivation: deblurring is harder than it looks

- "Undo blur" sounds like applying an inverse filter.
- In practice: noise, zeros/near-zeros in frequency response, instability.
- We need *stability* + *plausible images*.

Forward model (DSP viewpoint)

Model the image as a 2D discrete signal $x[i, j]$.

$$b = k * x + n$$

- b: observed blurred (and noisy) image
- k: PSF / blur kernel
- $*$: 2D convolution
- n: additive noise

Fourier-domain view: why inversion amplifies noise

Assuming circular convolution (periodic boundary), Fourier diagonalizes convolution:

$$\widehat{b}(\omega) = \widehat{k}(\omega)\,\widehat{x}(\omega) + \widehat{n}(\omega)$$

Naive inverse filter:

$$\widehat{x}(\omega) = \frac{\widehat{b}(\omega)}{\widehat{k}(\omega)}$$

Ill-conditioning and ill-posedness

- Deblurring is often ill-conditioned: small singular values of the blur operator.
- Practical symptoms:
  - small changes in b (noise) cause large changes in recovered x
  - ringing artifacts
  - over-amplified high frequencies

Regularized formulation: trade data fit vs prior

We replace least-squares only:

$$\min_x \ \frac{1}{2}\|Kx - b\|_2^2$$

by a regularized problem:

$$\min_x \ \frac{1}{2}\|Kx - b\|_2^2 + \lambda R(x)$$

- K: convolution operator for kernel k
- $R(x)$: prior / penalty enforcing plausible images
- $\lambda > 0$: tradeoff (fit vs smoothness/structure)

Classic option: Tikhonov (quadratic smoothing)

A common choice:

$$R(x) = \|\nabla x\|_2^2$$

- Equivalent to attenuating high frequencies.
- Often over-smooths edges (edges are high-frequency content).

Edge-preserving option: Total Variation (TV)

TV penalty:

$$R(x) = \|\nabla x\|_1$$

- Promotes piecewise-constant images with sharp edges.
- Known artifact: *staircasing* (flat plateaus).

Why consider second-order (Hessian) regularization?

- First derivatives capture edges, but can cause staircasing.
- Second derivatives capture *curvature*: how gradients change.
- Hessian-based penalties can encourage:
  - smooth regions remain flat (small curvature)
  - edges remain edges but with more natural transitions
  - reduced staircasing

Preview: what you will implement

We will solve:

$$\min_x \frac{1}{2} \|Kx - b\|_2^2 + \lambda \sum_i R(H(x)_i)$$

- $H(x)_i$: 2x2 Hessian matrix at pixel i
- $R(M)$: a mixture of **nuclear** and **Frobenius** norms of M

Hessian in 2D: local curvature of an image surface

Treat image as a surface $x(i, j)$.

$$H(x) = \begin{bmatrix} x_{xx} & x_{xy} \\ x_{xy} & x_{yy} \end{bmatrix}$$

- x_{xx}: curvature along horizontal direction
- x_{yy}: curvature along vertical direction
- x_{xy}: mixed curvature

Discrete second-derivative filters

Finite differences (periodic boundaries):

$$x_{xx} \approx [1, -2, 1] * x, \qquad x_{yy} \approx [1; -2; 1] * x$$

Mixed derivative (one common stencil):

$$x_{xy}[i, j] \approx \frac{x[i+1, j+1] - x[i+1, j-1] - x[i-1, j+1] + x[i-1, j-1]}{4}$$

Geometric meaning via eigenvalues/singular values

For a symmetric 2x2 Hessian at a pixel:

- eigenvalues (or singular values) quantify curvature magnitudes
- rough classification:
  - both small $\Rightarrow$ flat
  - one large, one small $\Rightarrow$ edge/filament
  - both large $\Rightarrow$ corner/blob-like curvature

Spectral norms of a 2x2 matrix

Let singular values of M be $s_1, s_2 \geq 0$.

$$\|M\|_* = s_1 + s_2 \qquad \text{(nuclear norm)}$$

$$\|M\|_F = \sqrt{s_1^2 + s_2^2} \qquad \text{(Frobenius norm)}$$

Why nuclear norm on the Hessian? (rank bias)

- Nuclear norm $\|M\|_* = s_1 + s_2$ encourages *low-rank* matrices.
- For 2x2: low-rank means often $s_2 \approx 0$.
- Interpretation: curvature mainly in *one direction* (edge-like).

Why Frobenius norm on the Hessian?

- Frobenius $\|M\|_F = \sqrt{s_1^2 + s_2^2}$ measures total curvature energy.
- More "isotropic": penalizes overall magnitude regardless of rank.
- Often yields smoother, more conservative regularization.

Mixed penalty: interpolate between two behaviors

Define per-pixel penalty:

$$R(M) = rw \, \|M\|_* + (1 - rw) \, \|M\|_F \,, \qquad rw \in [0, 1]$$

- $rw = 1$: pure nuclear (strong rank bias)
- $rw = 0$: pure Frobenius (energy-only)
- intermediate: mix of "structure" + "stability"

Full variational problem

We solve:

$$\min_x \frac{1}{2} \|Kx - b\|_2^2 + \lambda \sum_i R(H(x)_i)$$

- Large-scale optimization (one variable per pixel).
- Non-smooth due to nuclear norm and Frobenius norm.
- We need an algorithm designed for such composite objectives.

Splitting trick: introduce an auxiliary variable

Let

$$Z = Hx$$

Rewrite:

$$\min_{x,Z} \ \frac{1}{2}\|Kx - b\|_2^2 + \lambda \sum_i R(Z_i) \quad \text{s.t.} \quad Z = Hx$$

ADMM: three-step iteration

ADMM alternates:

1. x-update: minimize w.r.t. x (quadratic; solved in FFT domain)
2. Z-update: minimize w.r.t. Z (pixelwise proximal operator)
3. dual update: update scaled multiplier U

Augmented Lagrangian (scaled form)

Scaled augmented Lagrangian:

$$\mathcal{L}_\rho(x, Z, U) = \frac{1}{2}\|Kx - b\|_2^2 + \lambda \sum_i R(Z_i) + \frac{\rho}{2}\|Hx - Z + U\|_F^2 - \frac{\rho}{2}\|U\|_F^2$$

x-update: quadratic solve (FFT)

$$x^{k+1} = \arg\min_x \ \frac{1}{2}\|Kx - b\|_2^2 + \frac{\rho}{2}\left\|Hx - Z^k + U^k\right\|_F^2$$

Under periodic boundary conditions:

- K, H are convolutions $\Rightarrow$ diagonal in Fourier domain
- solve by elementwise division in frequency domain

Z-update: proximal operator (this is the key step)

$$Z^{k+1} = \arg\min_{Z} \ \lambda \sum_i R(Z_i) + \frac{\rho}{2} \left\| Z - (Hx^{k+1} + U^k) \right\|_F^2$$

Define

$$V = Hx^{k+1} + U^k, \qquad \tau = \lambda/\rho$$

Then:

$$Z^{k+1} = \mathrm{prox}_{\tau R}(V) \quad \text{(applied per pixel)}$$

Why does prox reduce to singular values?

Key fact: $R(M)$ depends only on singular values (unitarily invariant). If $V = U\ \mathrm{diag}(s)\, V^T$ (SVD), then

$$\mathrm{prox}_{\tau R}(V) = U\ \mathrm{diag}(t)\, V^T$$

where t solves a 2D vector proximal problem on s.

Singular-value prox problem

Let $s = (s_1, s_2)$ be singular values of V. We need:

$$t = \arg\min_{t \geq 0} \frac{1}{2} \|t - s\|_2^2 + \tau \Big(rw \|t\|_1 + (1 - rw) \|t\|_2 \Big)$$

Interpretation:

- $\|t\|_1$ promotes sparsity in singular values (rank bias)
- $\|t\|_2$ shrinks energy of singular values (magnitude control)

Closed form: soft-threshold then vector shrink

Define thresholds:

$$\alpha = \tau\, rw, \qquad \beta = \tau(1 - rw)$$

Step 1 (soft threshold each singular value):

$$u = \max(s - \alpha,\, 0) \quad \text{(elementwise)}$$

Step 2 (group shrink on vector u):

$$t = \max\left(1 - \frac{\beta}{\|u\|_2},\, 0\right) u$$

Reconstruction from shrunk singular values

Given $V = U\,\mathrm{diag}(s)V^T$ and computed t:

$$Z = U\,\mathrm{diag}(t)V^T$$

Implementation trick (no explicit diag):

$$Z = (U \odot t)\,V^T$$

(where $U \odot t$ means multiply each column of U by the corresponding entry of t).

Dual update

Scaled dual update:

$$U^{k+1} = U^k + (Hx^{k+1} - Z^{k+1})$$

Stopping criteria

We monitor:

$$r^k = Hx^k - Z^k \quad \text{(primal residual)}$$

$$s^k = \rho(Z^k - Z^{k-1}) \quad \text{(dual residual)}$$

Stop when both are below tolerances (already coded).

What you must implement (only one function)

Implement:

$$Z = \mathrm{prox}_{\tau[rw\|\cdot\|_* + (1-rw)\|\cdot\|_F]}(V)$$

in code as:

```
prox_mixed_nuclear_frob(Z, tau, rw)
```

Inputs:

- batch of 2x2 matrices (one per pixel)
- scalars τ and rw

Output:

- same-shaped batch of 2x2 matrices

Lab checklist: required behavior tests

Your function must satisfy:

- $\tau = 0$: output equals input
- $rw = 1$: equals nuclear prox (soft-threshold singular values by τ)
- $rw = 0$: equals Frobenius prox (shrink matrix magnitude by Frobenius norm)
- shape preservation: `(...,2,2)` $\rightarrow$ `(...,2,2)`

Debug strategy (recommended)

Start small:

- Test on a single diagonal matrix $V = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$
- Singular values are $|a|, |b|$ (easy to verify)
- Compare your prox output against manual shrink steps

Then test on random 2x2 matrices:

- Compare with a slow loop version (pixelwise SVD) before vectorizing

Experiments: explore the effect of rw

Run deblurring with:

$$rw \in \{0,\ 0.5,\ 1\}$$

Observe:

- edge sharpness
- smoothness in flat regions
- artifacts (ringing / texture loss)

Parameter intuition: λ and ρ

- λ controls *strength of regularization*
  - too small: noisy / unstable
  - too large: overly smooth / loss of detail
- ρ is an ADMM parameter (convergence speed/behavior)
  - typically tuned for stable residual decrease

Summary: what you learned

- Deblurring is ill-posed; regularization is essential.
- Hessian penalizes curvature; spectral norms use singular values.
- ADMM splitting makes the problem tractable.
- Your job: implement **one proximal operator** based on SVD + shrinkage.

Quick reference formulas

Mixed penalty

$$R(M) = rw \left\|M\right\|_* + (1 - rw) \left\|M\right\|_F$$

Singular-value prox

$$\alpha = \tau rw, \;\; \beta = \tau(1 - rw)$$
$$u = \max(s - \alpha, 0)$$
$$t = \max(1 - \beta / \left\|u\right\|_2, 0)\, u$$
$$Z = U \operatorname{diag}(t) V^T$$